# Supplementary Material: The MOND Interpolating Function from the Petz Recovery Bound

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We derive the MOND interpolating function $\mu(x) = 1 - e^{-\sqrt{x}}$ directly from the Petz recovery fidelity bound $F \geq e^{-\Sigma/2}$ established in Paper 1, with the identification $\Sigma_{\rm gal} = 2\sqrt{x}$ where $x = g_{\rm bar}/a_0$. This provides an information-theoretic origin for the transition between Newtonian and deep-MOND regimes: the interpolating function *is* the Petz recovery failure $\tau = 1 - F$. We verify that both asymptotic limits are correctly recovered, compare with standard interpolating functions used in the literature, and identify the remaining conjecture that requires microscopic justification.

## I. THE PETZ RECOVERY BOUND

We recall the central result of Paper 1 [1]. For any completely positive trace-preserving (CPTP) channel $\mathcal{E}$ with entropy production

$$\Sigma \;=\; D\big(\rho \,\big\|\, \mathcal{R}_{\mathcal{E}}[\mathcal{E}(\rho)]\big) \geq 0\,, \tag{1}$$

the Petz recovery fidelity satisfies the Junge–Renner–Sutter–Wilde–Winter (JRSWW) bound [2]

$$F(\rho,\, \mathcal{R}_{\mathcal{E}}[\mathcal{E}(\rho)]) \;\geq\; e^{-\Sigma/2}\,. \tag{2}$$

Defining the temporal asymmetry parameter as the Petz recovery failure,

$$\tau \;=\; 1 - F\,, \tag{3}$$

the bound (2) yields an upper envelope

$$\tau_{\rm bound} \;=\; 1 - e^{-\Sigma/2}\,. \tag{4}$$

This is a universal, model-independent inequality: any physical process with entropy production $\Sigma$ has its retrodictability failure bounded by Eq. (4). In Paper 1 we showed that this single equation, evaluated in different limits, reproduces the Crooks fluctuation theorem, Landauer's principle, and the gravitational Tolman–Oppenheimer–Volkoff redshift.

## II. GALACTIC IDENTIFICATION

At galactic scales, the gravitational field produces an entropy associated with the deviation of the observed acceleration from the Newtonian prediction. We define the standard MOND ratio

$$x \;\equiv\; \frac{g_{\rm bar}}{a_0}\,, \tag{5}$$

where $g_{\rm bar} = GM_{\rm bar}/r^2$ is the Newtonian baryonic acceleration and $a_0 \approx 1.2 \times 10^{-10}$ m/s$^2$ is the MOND acceleration scale.

[ASSUMED] We identify the galactic gravitational entropy production as

$$\boxed{\Sigma_{\rm gal} \;=\; 2\sqrt{x} \;=\; 2\sqrt{\frac{g_{\rm bar}}{a_0}}} \tag{6}$$

The physical motivation is as follows:

- In the *deep-MOND* regime ($x \to 0$): the gravitational field deviates maximally from Newton. $\Sigma_{\rm gal} \to 0$, reflecting that the Petz map can almost fully reconstruct the Newtonian state—the "retrodiction cost" per unit acceleration is small when there is almost no baryonic signal.
- In the *Newtonian* regime ($x \to \infty$): $\Sigma_{\rm gal} \to \infty$, so $\tau \to 1$ (saturation), meaning that the Newtonian dynamics is fully "irreversible" at the information-theoretic level—no MOND correction is needed.

The $\sqrt{x}$ form is the unique power law $x^\alpha$ that simultaneously gives $\mu \to 1$ as $x \to \infty$ (automatic for any $\alpha > 0$) and $\mu \to \sqrt{x}$ as $x \to 0$ (requires $\alpha = 1/2$; see Sec. IV).

## III. DERIVATION OF THE INTERPOLATING FUNCTION

The MOND interpolating function $\mu(x)$ relates the observed gravitational acceleration $g_{\rm obs}$ to the baryonic Newtonian prediction:

$$g_{\rm obs} \;=\; \frac{g_{\rm bar}}{\mu(x)}\,. \tag{7}$$

In the $\tau$ framework, the fractional deviation of gravity from Newton *is* the Petz recovery failure: the inability to reconstruct the Newtonian gravitational state from the observed state quantifies precisely how much "extra" gravity is present. We therefore identify

$$\mu(x) \;=\; \tau_{\rm bound} \;=\; 1 - e^{-\Sigma_{\rm gal}/2}\,. \tag{8}$$

Substituting the galactic identification (6):

$$\boxed{\mu(x) \;=\; 1 - e^{-\sqrt{x}}} \tag{9}$$

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This is the main result. The MOND interpolating function arises as the *saturated Petz recovery bound*—the boundary case where the gravitational channel is maximally efficient at converting entropy production into retrodiction failure. Note that:

1. No Boltzmann weights or thermal partition functions appear; this is a purely information-theoretic derivation.

2. The only inputs are the JRSWW bound (2) from quantum information theory and the identification (6) of the galactic entropy production.

3. The Crooks fluctuation theorem is *not* invoked. The derivation lives entirely within the Petz recovery framework.

## IV. ASYMPTOTIC VERIFICATION

**Newtonian limit** ($x \to \infty$).—As $x \to \infty$, $e^{-\sqrt{x}} \to 0$, hence

$$\mu(x) \;\to\; 1\,, \tag{10}$$

recovering $g_{\rm obs} = g_{\rm bar}$: standard Newtonian gravity. ✓

**Deep-MOND limit** ($x \to 0$).—Expanding to leading order,

$$\mu(x) = 1 - e^{-\sqrt{x}} = \sqrt{x} - \frac{x}{2} + \frac{x^{3/2}}{6} - \cdots \;\approx\; \sqrt{x}\,. \tag{11}$$

Substituting into Eq. (7):

$$g_{\rm obs} \;=\; \frac{g_{\rm bar}}{\sqrt{g_{\rm bar}/a_0}} \;=\; \sqrt{g_{\rm bar}\, a_0}\,. \tag{12}$$

For a circular orbit, $g_{\rm obs} = V^2/r$ and $g_{\rm bar} = GM/r^2$, giving

$$V^4 \;=\; GM\, a_0\,, \tag{13}$$

the Baryonic Tully–Fisher Relation (BTFR) with zero intrinsic scatter. ✓

**Transition region** ($x \sim 1$).—At $x = 1$: $\mu(1) = 1 - e^{-1} \approx 0.632$. The transition from deep-MOND to Newtonian occurs smoothly over approximately one decade in $x$, centered near $x \sim 1$.

## V. COMPARISON WITH OTHER INTERPOLATING FUNCTIONS

Table I compares the Petz-derived interpolating function with the standard choices in the MOND literature.

Several observations:

1. The Petz function (9) is *identical in functional form* to the empirical fit obtained by McGaugh, Lelli, and Schombert [4] from the SPARC Radial Acceleration Relation (RAR) data. This is a non-trivial consistency check: the information-theoretic derivation independently produces the same function that best fits 2693 data points from 153 galaxies.

2. The "simple" and "standard" interpolating functions have $\mu \sim x$ as $x \to 0$, which gives $g_{\rm obs} \sim g_{\rm bar}/x = a_0$—a constant acceleration floor. By contrast, the Petz function has $\mu \sim \sqrt{x}$, giving the physically correct $g_{\rm obs} = \sqrt{g_{\rm bar} a_0}$.

3. The transition sharpness differs: the Petz/McGaugh function transitions more gradually than the "standard" function ($\mu(1) = 0.632$ vs. 0.707) but more sharply than the "simple" function (0.632 vs. 0.500).

4. Since the Petz function matches the McGaugh RAR fit exactly, no additional fitting is needed against SPARC data. The agreement with observations is inherited directly.

TABLE I. Comparison of MOND interpolating functions. $x = g_{\rm bar}/a_0$.

| Name | $\mu(x)$ | $\mu(1)$ | $x \to 0$ |
|---|---|---|---|
| Petz (this work) | $1 - e^{-\sqrt{x}}$ | 0.632 | $\sqrt{x}$ |
| Simple [5] | $\dfrac{x}{1+x}$ | 0.500 | $x$ |
| Standard [6] | $\dfrac{x}{\sqrt{1+x^2}}$ | 0.707 | $x$ |
| McGaugh RAR [4] | $1 - e^{-\sqrt{x}}$ | 0.632 | $\sqrt{x}$ |

## VI. WHAT REMAINS CONJECTURAL

The derivation rests on one non-trivial identification:

$$\Sigma_{\rm gal} = 2\sqrt{x}\,. \tag{6}$$

This is [ASSUMED], not [DERIVED]. We offer three supporting arguments:

**(i) Dimensional consistency.** $\Sigma$ must be dimensionless. The combination $g_{\rm bar}/a_0$ is dimensionless, and its square root is the simplest function thereof. The prefactor 2 is fixed by matching the JRSWW bound normalization (2).

**(ii) Asymptotic uniqueness.** Consider the general ansatz $\Sigma_{\rm gal} = 2\,x^\alpha$. Then $\mu = 1 - e^{-x^\alpha}$. The deep-MOND limit $\mu \approx x^\alpha$ must equal $\sqrt{x}$ to reproduce the BTFR (13), forcing $\alpha = 1/2$. No other power law works.

**(iii) Connection to the cosmological sector.** If $\mu_0 = H_0/c$ (as suggested by Paper 3's cosmological analysis) and $a_0 = cH_0/(2\pi)$ (Milgrom's coincidence), then

$$\Sigma_{\rm gal} = 2\sqrt{\frac{g_{\rm bar}}{c^2\mu_0/(2\pi)}}\,, \tag{14}$$

linking the galactic entropy production to the same parameter $\mu_0$ that governs the cosmological sector. This suggests a unified microscopic origin, though the derivation of $\Sigma_{\rm gal} = 2\sqrt{x}$ from first principles remains an open problem.

A microscopic derivation would require computing the quantum relative entropy $D(\rho_{\rm obs}\|\rho_{\rm Newton})$ for the gravitational field in the weak-field, quasi-static limit. We leave this to future work.

## VII. SUMMARY

The derivation chain is:

$$\underbrace{F \geq e^{-\Sigma/2}}_{\text{JRSWW bound}} \xrightarrow{\Sigma_{\rm gal}=2\sqrt{x}} \underbrace{\mu(x) = 1 - e^{-\sqrt{x}}}_{\text{MOND interpolation}} \tag{15}$$

The Petz recovery bound, combined with a single identification of the galactic entropy production, yields the MOND interpolating function that coincides with the best empirical fit to the Radial Acceleration Relation. The derivation uses no Boltzmann weights, no Crooks theorem, and no partition functions—only the information-theoretic structure of quantum channels.

## ACKNOWLEDGMENTS

This work builds on the $\tau$ framework established in Paper 1 [1].

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